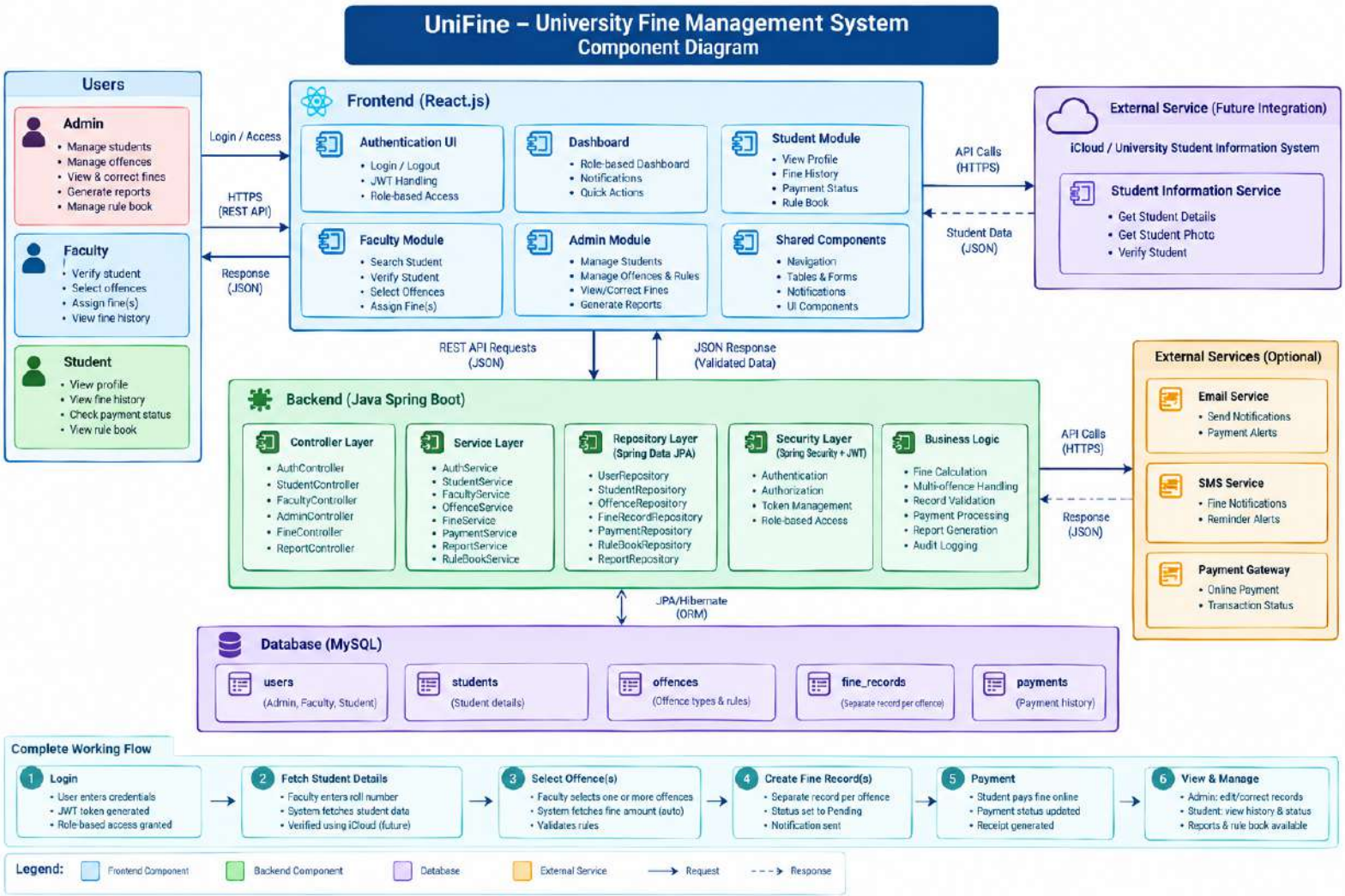
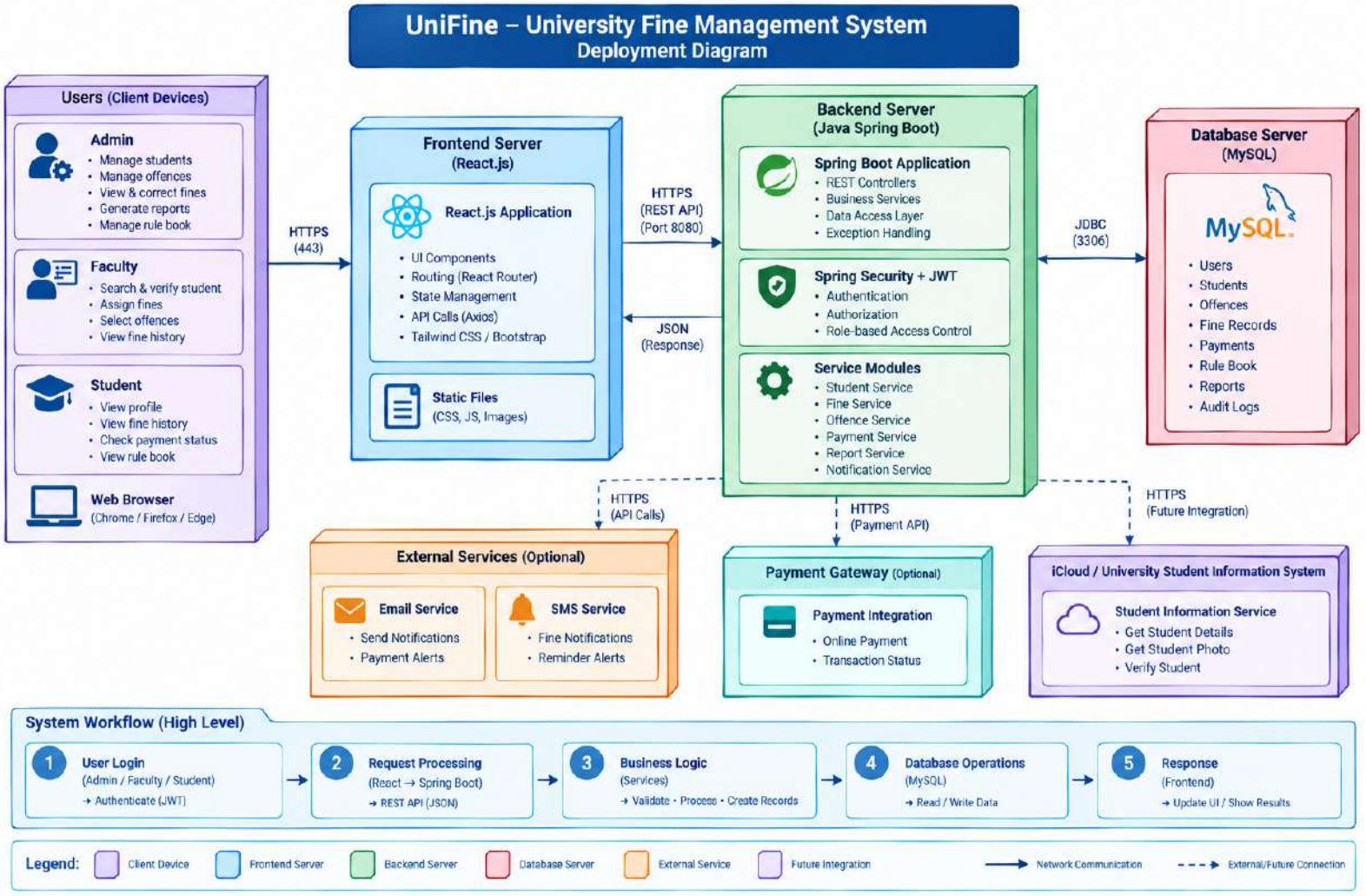
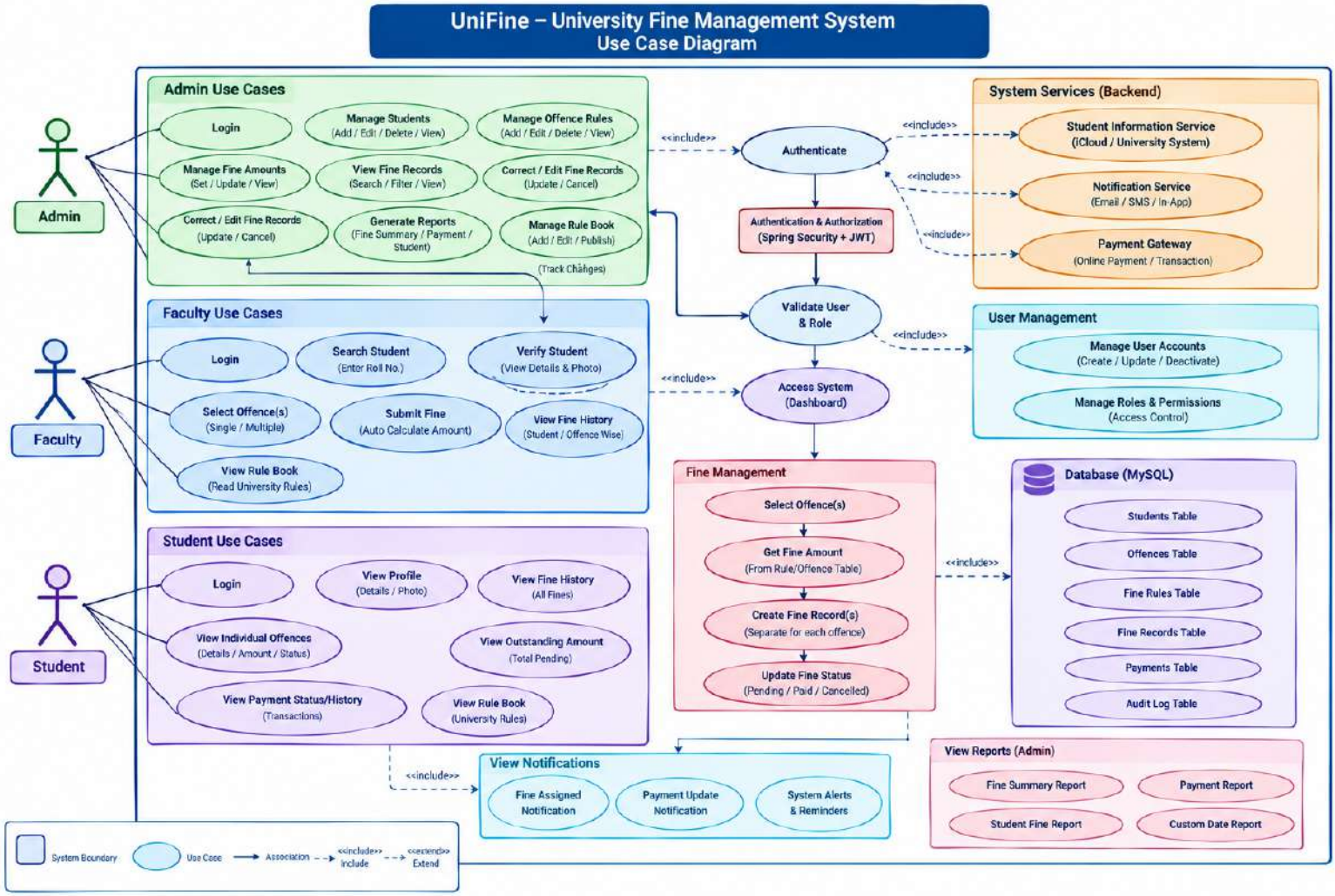
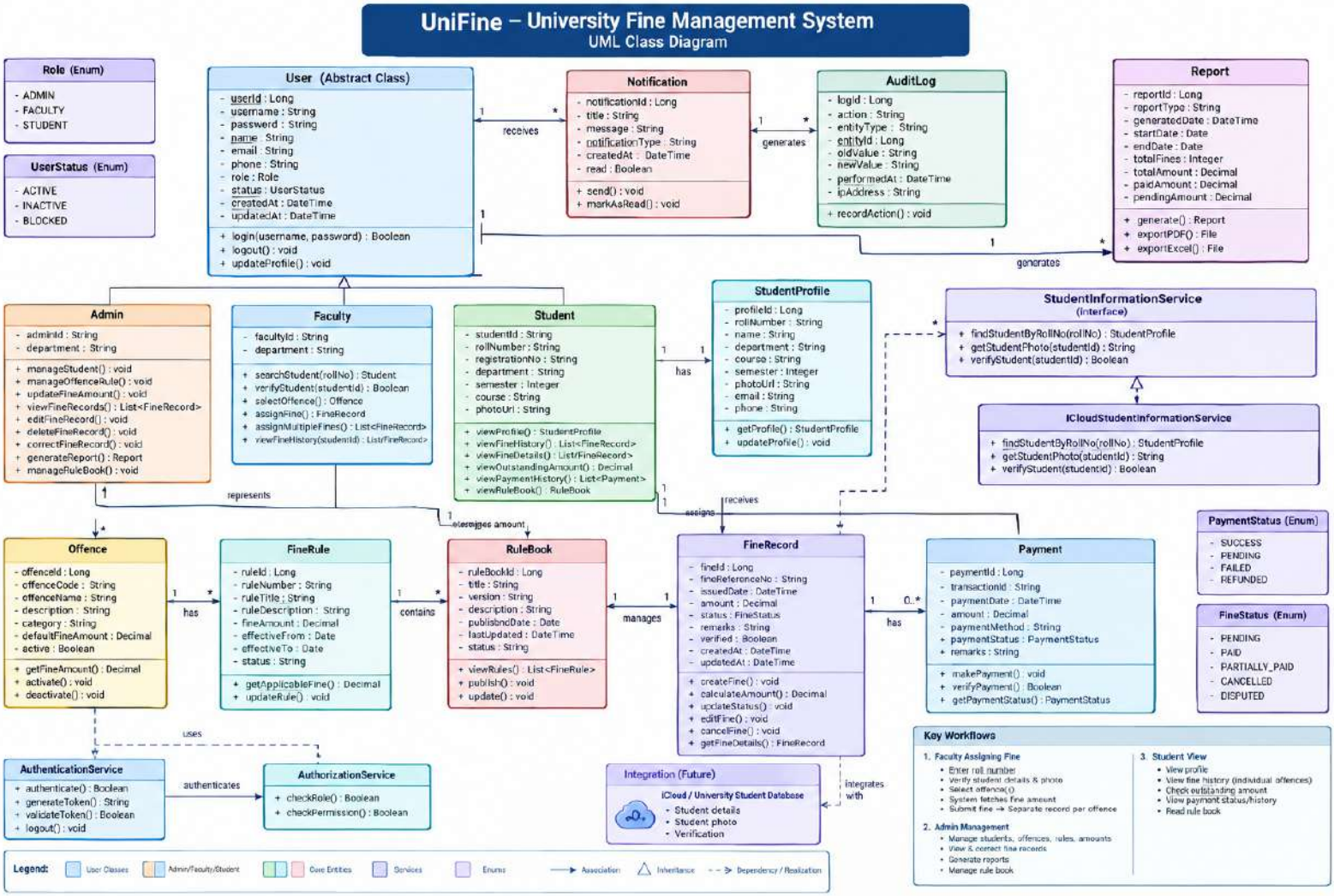

UniFine – University Fine Management System
Component Diagram
Users
Admin
Manage students
Manage offences
View & correct fines
Generate reports
Manage rule book
Faculty
Verify student
Select offences
Assign fine(s)
View fine history
Student
View profile
View fine history
Check payment status
View rule book
Login / Access
HTTPS (REST API)
Response (JSON)
Frontend (React.js)
Authentication UI
Login / Logout
JWT Handling
Role-based Access
Dashboard
Role-based Dashboard
Notifications
Quick Actions
Student Module
View Profile
Fine History
Payment Status
Rule Book
Faculty Module
Search Student
Verify Student
Select Offences
Assign Fine(s)
Admin Module
Manage Students
Manage Offences & Rules
View/Correct Fines
Generate Reports
Shared Components
Navigation
Tables & Forms
Notifications
UI Components
API Calls (HTTPS)
Student Data (JSON)
External Service (Future Integration)
iCloud / University Student Information System
Student Information Service
Get Student Details
Get Student Photo
Verify Student
REST API Requests (JSON)
JSON Response (Validated Data)
Backend (Java Spring Boot)
Controller Layer
AuthController
StudentController
FacultyController
AdminController
FineController
ReportController
Service Layer
AuthService
StudentService
FacultyService
OffenceService
FineService
PaymentService
ReportService
RuleBookService
Repository Layer (Spring Data JPA)
UserRepository
StudentRepository
OffenceRepository
FineRecordRepository
PaymentRepository
RuleBookRepository
ReportRepository
Security Layer (Spring Security + JWT)
Authentication
Authorization
Token Management
Role-based Access
Business Logic
Fine Calculation
Multi-offence Handling
Record Validation
Payment Processing
Report Generation
Audit Logging
API Calls (HTTPS)
Response (JSON)
External Services (Optional)
Email Service
Send Notifications
Payment Alerts
SMS Service
Fine Notifications
Reminder Alerts
Payment Gateway
Online Payment
Transaction Status
JPA/Hibernate (ORM)
Database (MySQL)
users
(Admin, Faculty, Student)
students
(Student details)
offences
(Offence types & rules)
fine_records
(Separate record per offence)
payments
(Payment history)
Complete Working Flow
1 Login
User enters credentials
JWT token generated
Role-based access granted
2 Fetch Student Details
Faculty enters roll number
System fetches student data
Verified using iCloud (future)
3 Select Offence(s)
Faculty selects one or more offences
System fetches fine amount (auto)
Validates rules
4 Create Fine Record(s)
Separate record per offence
Status set to Pending
Notification sent
5 Payment
Student pays fine online
Payment status updated
Receipt generated
6 View & Manage
Admin: edit/correct records
Student: view history & status
Reports & rule book available
Legend:
Frontend Component
Backend Component
Database
External Service
Request
Response


UniFine – University Fine Management System
Deployment Diagram
Users (Client Devices)
Admin
• Manage students
• Manage offences
• View & correct fines
• Generate reports
• Manage rule book
Faculty
• Search & verify student
• Assign fines
• Select offences
• View fine history
Student
• View profile
• View fine history
• Check payment status
• View rule book
Web Browser
(Chrome / Firefox / Edge)
HTTPS
(443)
Frontend Server
(React.js)
React.js Application
• UI Components
• Routing (React Router)
• State Management
• API Calls (Axios)
• Tailwind CSS / Bootstrap
Static Files
(CSS, JS, Images)
HTTPS
(REST API)
(Port 8080)
JSON
(Response)
Backend Server
(Java Spring Boot)
Spring Boot Application
• REST Controllers
• Business Services
• Data Access Layer
• Exception Handling
Spring Security + JWT
• Authentication
• Authorization
• Role-based Access Control
Service Modules
• Student Service
• Fine Service
• Offence Service
• Payment Service
• Report Service
• Notification Service
JDBC
(3306)
Database Server
(MySQL)
MySQL
• Users
• Students
• Offences
• Fine Records
• Payments
• Rule Book
• Reports
• Audit Logs
HTTPS
(API Calls)
HTTPS
(Payment API)
HTTPS
(Future Integration)
External Services (Optional)
Email Service
• Send Notifications
• Payment Alerts
SMS Service
• Fine Notifications
• Reminder Alerts
Payment Gateway (Optional)
Payment Integration
• Online Payment
• Transaction Status
iCloud / University Student Information System
Student Information Service
• Get Student Details
• Get Student Photo
• Verify Student
System Workflow (High Level)
1 User Login
(Admin / Faculty / Student)
→ Authenticate (JWT)
2 Request Processing
(React → Spring Boot)
→ REST API (JSON)
3 Business Logic
(Services)
→ Validate • Process • Create Records
4 Database Operations
(MySQL)
→ Read / Write Data
5 Response
(Frontend)
→ Update UI / Show Results
Legend:
Client Device
Frontend Server
Backend Server
Database Server
External Service
Future Integration
Network Communication
External/Future Connection


UniFine – University Fine Management System
Use Case Diagram
Admin
Faculty
Student
Admin Use Cases
Login
Manage Students
(Add / Edit / Delete / View)
Manage Offence Rules
(Add / Edit / Delete / View)
Manage Fine Amounts
(Set / Update / View)
View Fine Records
(Search / Filter / View)
Correct / Edit Fine Records
(Update / Cancel)
Correct / Edit Fine Records
(Update / Cancel)
Generate Reports
(Fine Summary / Payment / Student)
Manage Rule Book
(Add / Edit / Publish)
(Track Changes)
Faculty Use Cases
Login
Search Student
(Enter Roll No.)
Verify Student
(View Details & Photo)
Select Offence(s)
(Single / Multiple)
Submit Fine
(Auto Calculate Amount)
View Fine History
(Student / Offence Wise)
View Rule Book
(Read University Rules)
Student Use Cases
Login
View Profile
(Details / Photo)
View Fine History
(All Fines)
View Individual Offences
(Details / Amount / Status)
View Outstanding Amount
(Total Pending)
View Payment Status/History
(Transactions)
View Rule Book
(University Rules)
<<include>>
Authenticate
Authentication & Authorization
(Spring Security + JWT)
Validate User
& Role
Access System
(Dashboard)
System Services (Backend)
Student Information Service
(iCloud / University System)
Notification Service
(Email / SMS / In-App)
Payment Gateway
(Online Payment / Transaction)
User Management
Manage User Accounts
(Create / Update / Deactivate)
Manage Roles & Permissions
(Access Control)
Fine Management
Select Offence(s)
Get Fine Amount
(From Rule/Offence Table)
Create Fine Record(s)
(Separate for each offence)
Update Fine Status
(Pending / Paid / Cancelled)
Database (MySQL)
Students Table
Offences Table
Fine Rules Table
Fine Records Table
Payments Table
Audit Log Table
View Notifications
Fine Assigned Notification
Payment Update Notification
System Alerts & Reminders
View Reports (Admin)
Fine Summary Report
Payment Report
Student Fine Report
Custom Date Report
System Boundary
Use Case
Association
<<include>>
include
<<extend>>
Extend


UniFine – University Fine Management System
UML Class Diagram
Role (Enum)
- ADMIN
- FACULTY
- STUDENT
UserStatus (Enum)
- ACTIVE
- INACTIVE
- BLOCKED
User (Abstract Class)
- userId : Long
- username : String
- password : String
- name : String
- email : String
- phone : String
- role : Role
- status : UserStatus
- createdAt : DateTime
- updatedAt : DateTime
+ login(username, password) : Boolean
+ logout() : void
+ updateProfile() : void
receives
Notification
- notificationId : Long
- title : String
- message : String
- notificationType : String
- createdAt : DateTime
- read : Boolean
+ send() : void
+ markAsRead() : void
generates
AuditLog
- logId : Long
- action : String
- entityType : String
- entityId : Long
- oldValue : String
- newValue : String
- performedAt : DateTime
- ipAddress : String
+ recordAction() : void
Report
- reportId : Long
- reportType : String
- generatedDate : DateTime
- startDate : Date
- endDate : Date
- totalFines : Integer
- totalAmount : Decimal
- paidAmount : Decimal
- pendingAmount : Decimal
+ generate() : Report
+ exportPDF() : File
+ exportExcel() : File
generates
Admin
- adminId : String
- department : String
+ manageStudent() : void
+ manageOffenceRule() : void
+ updateFineAmount() : void
+ viewFineRecords() : List<FineRecord>
+ editFineRecord() : void
+ deleteFineRecord() : void
+ correctFineRecord() : void
+ generateReport() : Report
+ manageRuleBook() : void
Faculty
- facultyId : String
- department : String
+ searchStudent(rollNo) : Student
+ verifyStudent(studentId) : Boolean
+ selectOffence() : Offence
+ assignFine() : FineRecord
+ assignMultipleFines() : List<FineRecord>
+ viewFineHistory(studentId) : List<FineRecord>
Student
- studentId : String
- rollNumber : String
- registrationNo : String
- department : String
- semester : Integer
- course : String
- photoUrl : String
+ viewProfile() : StudentProfile
+ viewFineHistory() : List<FineRecord>
+ viewFineDetails() : List<FineRecord>
+ viewOutstandingAmount() : Decimal
+ viewPaymentHistory() : List<Payment>
+ viewRuleBook() : RuleBook
has
StudentProfile
- profileId : Long
- rollNumber : String
- name : String
- department : String
- course : String
- semester : Integer
- photoUrl : String
- email : String
- phone : String
+ getProfile() : StudentProfile
+ updateProfile() : void
StudentInformationService
(Interface)
+ findStudentByRollNo(rollNo) : StudentProfile
+ getStudentPhoto(studentId) : String
+ verifyStudent(studentId) : Boolean
ICloudStudentInformationService
+ findStudentByRollNo(rollNo) : StudentProfile
+ getStudentPhoto(studentId) : String
+ verifyStudent(studentId) : Boolean
represents
assigns
receives
charges amount
Offence
- offenceId : Long
- offenceCode : String
- offenceName : String
- description : String
- category : String
- defaultFineAmount : Decimal
- active : Boolean
+ getFineAmount() : Decimal
+ activate() : void
+ deactivate() : void
has
FineRule
- ruleId : Long
- ruleNumber : String
- ruleTitle : String
- ruleDescription : String
- fineAmount : Decimal
- effectiveFrom : Date
- effectiveTo : Date
- status : String
+ getApplicableFine() : Decimal
+ updateRule() : void
contains
RuleBook
- ruleBookId : Long
- title : String
- version : String
- description : String
- publishedDate : Date
- lastUpdated : DateTime
- status : String
+ viewRules() : List<FineRule>
+ publish() : void
+ update() : void
manages
FineRecord
- fineId : Long
- fineReferenceNo : String
- issuedDate : DateTime
- amount : Decimal
- status : FineStatus
- remarks : String
- verified : Boolean
- createdAt : DateTime
- updatedAt : DateTime
+ createFine() : void
+ calculateAmount() : Decimal
+ updateStatus() : void
+ editFine() : void
+ cancelFine() : void
+ getFineDetails() : FineRecord
has
Payment
- paymentId : Long
- transactionId : String
- paymentDate : DateTime
- amount : Decimal
- paymentMethod : String
+ paymentStatus : PaymentStatus
+ remarks : String
+ makePayment() : void
+ verifyPayment() : Boolean
+ getPaymentStatus() : PaymentStatus
PaymentStatus (Enum)
- SUCCESS
- PENDING
- FAILED
- REFUNDED
FineStatus (Enum)
- PENDING
- PAID
- PARTIALLY_PAID
- CANCELLED
- DISPUTED
uses
AuthenticationService
+ authenticate() : Boolean
+ generateToken() : String
+ validateToken() : Boolean
+ logout() : void
authenticates
AuthorizationService
+ checkRole() : Boolean
+ checkPermission() : Boolean
Integration (Future)
ICloud / University Student Database
• Student details
• Student photo
• Verification
integrates with
Key Workflows
1. Faculty Assigning Fine
• Enter roll number
• Verify student details & photo
• Select offence(s)
• System fetches fine amount
• Submit fine → Separate record per offence
2. Admin Management
• Manage students, offences, rules, amounts
• View & correct fine records
• Generate reports
• Manage rule book
3. Student View
• View profile
• View fine history (individual offences)
• Check outstanding amount
• View payment status/history
• Read rule book
Legend:
User Classes
Admin/Faculty/Student
Core Entities
Services
Enums
Association
Inheritance
Dependency / Realization